AVRAMOV FOR LOCAL HOMOMORPHISMS: FINITELY MANY TOR CHARACTERIZE CI HOMOMORPHISMS

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Editorial note. The final step reads 'The proposition now follows from Lemma lemma-perfect-map-ci': the dictionary (J regular iff B CI via the presentation, finite tor dimension over R/I iff finitely many nonzero Tor) is implicit in the reduced setup; the completion reduction preceding it is fully displayed.

Allowed standard prerequisites (editorial).

- Embedding of complete local maps. more-algebra-lemma-embed-map-Noetherian-complete-local-rings (cross-chapter, named). Supplies the presentation diagram for the reduced statement.
- Three-way equivalence. lemma-perfect-map-ci (same chapter, named). The terminal engine: the reduced proposition is its special case.

Source reference. Original reference 00MB: focused-seeds/geometry/S9-dpa-avramov-map/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00MA: focused-seeds/geometry/S9-dpa-avramov-map/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 09PX: sources/stacks-dpa/pinned-original-source.tex, source lines 1836–1836. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07NN: focused-seeds/geometry/S9-dpa-avramov-map/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Proposition 0.1. *Let $A \to B$ be a local homomorphism of Noetherian local rings. Then the following are equivalent*

(1) *B is a complete intersection and $Tor_p^A(B, A/\mathfrak{m}_A)$ is nonzero for only finitely many p,*

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(2) *A is a complete intersection and $A^\wedge \to B^\wedge$ is a complete intersection homomorphism in the sense defined above.*

Proof. Let $F_\bullet \to A/\mathfrak{m}_A$ be a resolution by finite free A-modules. Observe that $\mathrm{Tor}_p^A(B, A/\mathfrak{m}_A)$ is the pth homology of the complex $F_\bullet \otimes_A B$. Let $F_\bullet^\wedge = F_\bullet \otimes_A A^\wedge$ be the completion. Then $F_\bullet^\wedge$ is a resolution of $A^\wedge/\mathfrak{m}_{A^\wedge}$ by finite free $A^\wedge$-modules (as $A \to A^\wedge$ is flat and completion on finite modules is exact, see Algebra, Lemmas 00MA and 00MB). It follows that

$$F_\bullet^\wedge \otimes_{A^\wedge} B^\wedge = F_\bullet \otimes_A B \otimes_B B^\wedge$$

By flatness of $B \to B^\wedge$ we conclude that

$$\mathrm{Tor}_p^{A^\wedge}(B^\wedge, A^\wedge/\mathfrak{m}_{A^\wedge}) = \mathrm{Tor}_p^A(B, A/\mathfrak{m}_A) \otimes_B B^\wedge$$

In this way we see that the condition in (1) on the local ring map $A \to B$ is equivalent to the same condition for the local ring map $A^\wedge \to B^\wedge$. Thus we may assume A and B are complete local Noetherian rings (since the other conditions are formulated in terms of the completions in any case).

Assume A and B are complete local Noetherian rings. Choose a diagram

$$\begin{array}{ccc} S & \longrightarrow & B \\ \uparrow & & \uparrow \\ R & \longrightarrow & A \end{array}$$

as in More on Algebra, Lemma 07NN. Let $I = \mathrm{Ker}(R \to A)$ and $J = \mathrm{Ker}(S \to B)$. The proposition now follows from Lemma 09PX. $\square$